

SUMMARY

Analyst with clinical experience and a strong foundation in healthcare operations, medical record management, data analytics, and process optimization. Skilled in Python, SQL, R, and healthcare data standards, with expertise in identifying trends, resolving discrepancies, and improving care delivery through data-driven insights. Experienced in working cross-functionally and passionate about streamlining healthcare workflows to support providers in delivering patient-centered care.

PROFESSIONAL EXPERIENCE

Indiana University Indianapolis

Indianapolis, IN

Research Data Analyst and Teaching Assistant

Jan 2024 – May 2025

- Queried OpenEMR using SQL in phpMyAdmin to analyze visit frequency and treatment patterns, revealing care trends.
- Mentored students in SQL query development and database management, encouraging collaboration across teams.
- Performed ETL on 300+ FAIR articles, built topic models, and visualized thematic gaps using VOS viewer for strategic planning.

Elimu Informatics

El Cerrito, CA

Clinical Informatics Intern

May 2024 – Jan 2025

- Developed a genomic decision support web application using FHIR APIs and Streamlit to help clinicians explore genomic-condition relationships, improving clinical decision-making.
- Automated variant interpretation and associated condition flagging using a Python script with FHIR Genomic Operations API, supported by ClinVar aggregation and SNOMED value set mapping, reducing manual review time.
- Created terminology translation API across SNOMED, ICD-10, and custom value sets, enhancing healthcare system interoperability.

Randolph County Caring Community Partnership

Moberly, MO

Decision Analyst Intern

Jun 2024 – Aug 2024

- Developed and validated clinical decision rules using CDC guidelines to enhance screening protocols and referral workflows.
- Designed Power BI dashboards with DAX for metrics tracking and reporting, enabling faster outreach and resource planning.
- Conducted literature reviews, authored reports for stakeholders and collaborated with cross-functional teams on protocol design.

Smile Essence Dental Clinic

MH, India

Health Data Analyst and Consultant Endodontist

May 2020 – Jul 2023

- Successfully managed dental practice operations and modernized patient record management by integrating a new Electronic Health Records (EHR) system, resulting in improved data accessibility.
- Performed SQL-based abstraction, Excel-based analysis and review of 1,000+ patient records to identify care trends and optimize appointment scheduling, increasing operational efficiency by 12%.
- Maintained clinical documentation and quality logs, supporting treatment progress tracking and regulatory compliance.

ACADEMIC PROJECTS

Hospital Readmission Analysis for Diabetic Patients | SQL, Python, Tableau

- Built a MySQL database of 80,000 patient records, performed data cleaning using Pandas and analyzed 35,523 patient records from 130 US hospitals (1999-2008) using SQL, identifying key factors influencing 30-day readmission for diabetic patients, and created interactive dashboards using Tableau to visualize trends, optimize hospital costs.

Wound Healing Trajectory Forecasting using Time Series Models | Python, Matplotlib, Time Series Analysis

- Led data cleaning and preprocessing of 200,000+ patient records, ensuring data quality, temporal alignment, and PHI compliance; assisted with IRB documentation and approvals; performed feature extraction and developed time series predictive models to forecast wound healing trajectories, supporting early identification of high-risk patients for timely clinical intervention.

Statistical Analysis of Factors associated with Chronic Kidney Disease | R, R Studio, Statistics

- Evaluated 400 records using R Studio, applying descriptive statistics, regression modeling, and categorical data analysis to identify significant chronic kidney disease correlations. Generated comprehensive visualizations using ggplot to communicate key findings.

Healthcare Diabetes Risk Prediction | Python, Machine Learning

- Developed predictive machine learning models for diabetes risk assessment, achieving 98% accuracy, applying advanced data preprocessing and statistical analysis techniques to identify critical predictive factors for early diagnostic intervention.

EDUCATION

Indiana University Indianapolis

MS in Health Informatics | GPA: 3.95

Aug 2023 - May 2025

Dr D Y Patil Vidyapeeth

Master of Dental Surgery | GPA: 3.6

May 2018 – Aug 2021

Maharashtra University of Health Sciences

Bachelor of Dental Surgery | GPA: 3.5

Sept 2011 – Sept 2016

TECHNICAL SKILLS

Programming Languages: Python (NumPy, Pandas, Matplotlib, Seaborn, PyTorch, Tensorflow), R Programming

Data Analysis Techniques: Machine Learning, Statistical Analysis, Time-Series Regression

Data Management Tools: JupyterHub, SQL (MySQL), phpMyAdmin, GitHub, REDCap, Microsoft Office (Excel, Outlook), VS Code

Data Visualization Tools: Tableau, Power BI, QlikView, Streamlit, VOS Viewer

Healthcare Standards: CPT, SNOMED, ICD, LOINC, FHIR, HL7

Regulatory Knowledge: Good Clinical Practice (GCP), HIPAA, Informed Consent, IRB